Wk 2 Supervision Problems 1

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A

Newton's law of cooling states that the rate an object cools is proportional to the temperature difference between the object and its surroundings. A 90°C cup of coffee is placed in a room maintained at 20°C. After 10 minutes, the coffee has cooled to 70°C.

1. Set up and solve an initial value problem for the temperature, $T(t)$, of the coffee using Newton's law of cooling.
2. Find the time at which the coffee reaches 40°C.

B

The displacement of a mass on a spring can be described by Newton's second law as follows:

$$m\frac{d^2}{dt^2}x(t) + c\frac{d}{dt}x(t) + kx(t) = 0$$

where m is the mass, c is the damping constant, and k is the spring constant. A mass of 1 kg is attached to a spring with constant $k = 13$ N/m. The system experiences damping with coefficient $c = 4$ Ns/m. At $t = 0$ the mass is displaced 0.5 m from equilibrium and released from rest.

1. Write down the initial value problem governing the displacement of the mass, $x(t)$, and find the complimentary function.
2. Solve the problem completely and determine the long-time behaviour of the system.